



2025

E22

BOOKLET NO.

128131

AGRICULTURE

Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

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P.T.O.

**SECTION – A**

Q1. Write the answers of following in **150** words **each**. **(5×10=50)**

- Q1.** (a) Define ecology and state different types of ecology. Explain in brief principles of agro-ecology. **10**
- Q1.** (b) Define soil conservation. Explain in brief methods of soil conservation. **10**
- Q1.** (c) Define Water Use Efficiency (WUE) and explain in brief criteria for scheduling irrigation. **10**
- Q1.** (d) What are the agro-climatic zones of India ? Explain in short its cropping pattern. **10**
- Q1.** (e) Define environmental pollution. Explain in brief different types of environmental pollution and its associated hazards. **10**

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- Q2.** (a) i) State and explain the principles of organic farming. **10**
ii) What are the advantages and limitations of organic farming ? **10**
- Q2.** (b) i) Explain commercial cultivation of suru/seasonal sugarcane on the following points. **10**
A) Soil and climate
B) Seed (set) and sowing
C) Fertilizer and irrigation management
D) Pest and disease management
E) Maturity signs, harvesting and yield
- ii) Describe improved production technology of pigeonpea on the following points. **10**
A) Soil and climate
B) Varieties
C) Fertilizer and weed management
D) Pest and disease management
E) Harvesting, yield and economic importance
- Q2.** (c) Explain in brief green house effect and global warming. **10**
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- Q3. (a)** i) Enlist advance tools for ecosystem analysis and describe the Remote Sensing (RS) and Geographic Information Systems (GIS). 10
ii) Write short note on international conventions and global initiatives on climate change. 10
- Q3. (b)** i) What are the characteristics of weed ? How are weeds disseminated ? 10
ii) What is weed ? Explain in brief different methods of weed control. 10
- Q3. (c)** Write in short methods to drain water from water logged soils. Explain in brief major irrigation projects in India (any five). 10
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- Q4. (a)** i) What are the methods of irrigation ? Describe in detail sprinkler and drip irrigation methods with their advantages and disadvantages. 10
ii) Define water harvesting. Explain in brief water harvesting. 10
- Q4. (b)** i) Define dry farming, dryland farming and dryland agriculture and explain in brief management practices of assured rainfall, medium rainfall and dry area of agriculture. 10
ii) What are the major constraints of dry farming ? Explain in brief mechanical and agronomic measures of soil and moisture conservation. 10
- Q4. (c)** What are the physical and social environmental factors of crop distribution and production ? Explain in brief. 10
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SECTION – B

- Q5.** Write the answers of following in **150 words each**. (5×10=50)
- Q5. (a)** i) Discuss the various ways in which people can participate in social forestry programmes. 6
ii) Write a short note on taungya system of agroforestry. 4
- Q5. (b)** State the Stoke's law with assumptions. Which specific processes are involve in soil formation ? Explain any two fundamental soil forming processes. 10
- Q5. (c)** What do you mean by farm management ? Write in brief scope, importance and characteristics of farm management. 10
- Q5. (d)** Discuss the role and importance of agricultural extension in the context of climate change in Maharashtra. 10
- Q5. (e)** Enlist major soil groups of India and describe in brief black soils and Laterite and Lateritic soils. 10
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- Q6. (a)** Give the Arnon's criteria for essentiality of nutrients. How the nutrients are classified based on quantity required by the plant ? Which approaches are suggest for fertilizer recommendations based on Soil Test Crop Response studies and explain any two approaches. **20**
- Q6. (b)** i) What are the characteristics of good farm plan ? What techniques will you adopt to solve the farm management problems ? **10**
ii) Describe in brief the role of cooperatives in agricultural economy of India. **10**
- Q6. (c)** i) Discuss the various vegetative propagation methods used in forestry. **6**
ii) Explain various products obtained from forests. **4**
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- Q7. (a)** What are biofertilizers ? Give the importance and classification of biofertilizers. Discuss the various means of nitrogen losses in soil. **20**
- Q7. (b)** i) What is the price fluctuations ? What are costs of price fluctuations for farmers ? **10**
ii) Write short note on Pradhan Mantri Fasal Bima Yojana. **10**
- Q7. (c)** Analyse the role of Krishi Vigyan Kendra (KVK's) in the changing scenario of Indian Agriculture. **10**
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- Q8. (a)** Give characteristics of salt affected soils. Explain in brief reclamation and management practices of saline soils. Discuss in short soil factors affecting green house gas emission. **20**
- Q8. (b)** Critically examine the Self-Help Group (SHG) approach for Rural Development in India. **20**
- Q8. (c)** Elaborate various methods of biodiversity conservation. **10**
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